Beyond the Basics: Using the New What-If for Deployment Stacks, Debugging, and AI-Assisted Bicep

Torrey Trahanovsky

Product Manager, Azure Deployments, Microsoft

NIC26

Before we start

Quick show of hands

- Who uses Bicep today?
- Who has run What-If?
- Who's lost sleep over What-If noise?
- Who uses Deployment Stacks?

This is a 300-level session

Bicep is Azure's native infrastructure as code language.

Good pre-reqs: deploying a Bicep file, and What-If.

Bicep docs	`aka.ms/bicep`
Learn Bicep	`aka.ms/learnbicep`
What-If	`aka.ms/whatif`

NIC26

Agenda

1 ~30 min

Stacks What-If

The new What-If for stacks

2 ~5 min

Debugging

Snapshot and lint

3 ~10 min

AI-assisted Bicep

Authoring resources as a stack

Time will be reserved for questions at the end, ~10 minutes.

NIC26

PART 1

The new What-If for Deployment Stacks

NIC26

What is What-If?

What-If compares the current state in Azure with your template to predict change.

Before you deploy

Run it on the template you're about to deploy.

Template vs. Azure

Each resource is checked against what's deployed now.

Predicted changes

What would be created or changed. Nothing gets deployed.

NIC26

Demo

Regular What-If, no change and then one change

Limitations of regular What-If

Noise

Differences you didn't make, like default values Azure adds, read-only properties and hidden secrets.

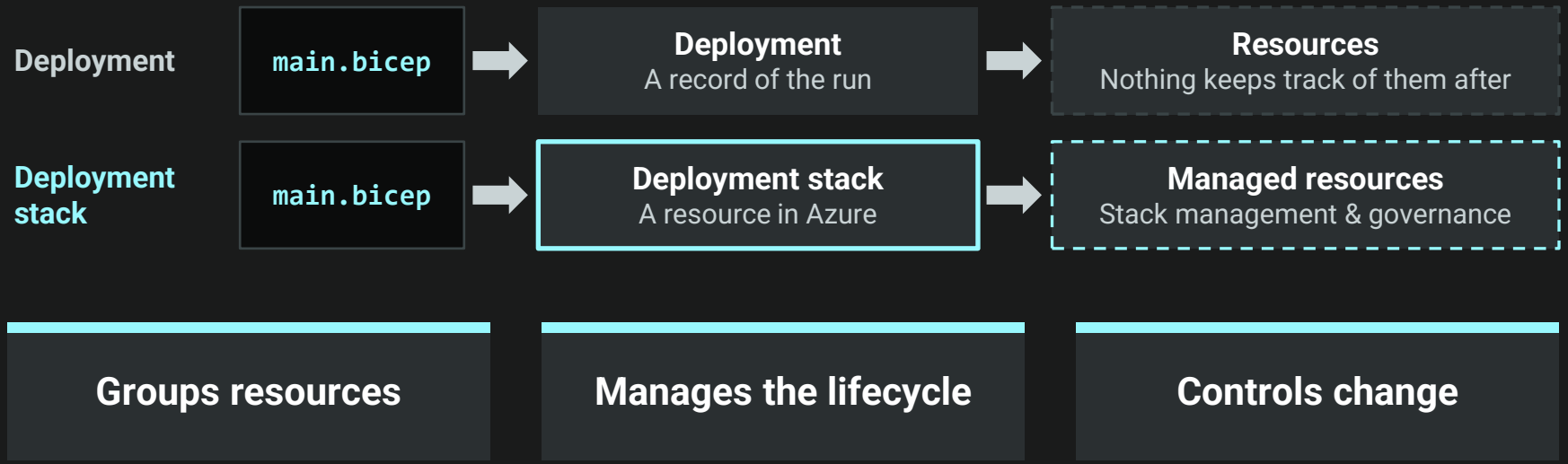
Removed resources

Take a resource out of the template, and What-If says Ignore.

No record

The result is only in your terminal or pipeline log.

What are Deployment Stacks?



Groups resources

Manages the lifecycle

Controls change

NIC26

Demo

Deploy it as a stack, then regular What-If again

What is Stacks What-If?

Stacks What-If predicts what a stack will change, before you create or update it.

Less noise

A lot of the noise gets filtered out.

What the stack does

Management and deny status, and detach or delete.

Saved as a resource

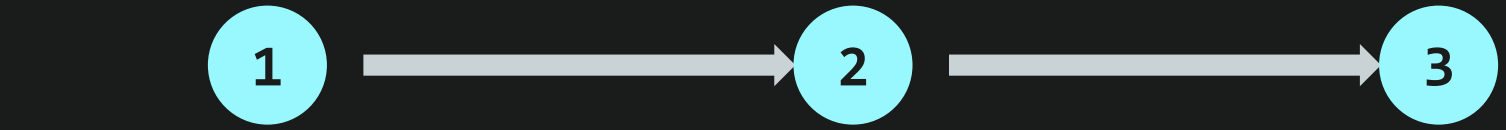
With a name you pick, kept for 1 to 3 hours.

NIC26

Demo

Stacks What-If on the same change

Noise reduction



You deploy or update the stack

Azure runs Stacks What-If and keeps it as a baseline

Your next What-If drops what was already in the baseline

Stacks not deployed since August need one deploy to get a baseline.

It filters a lot of noise, but not every difference, so still read the result. Feedback is welcome.

Reading the result

+ **Create** A new resource	~ **Modify** Properties change	= **NoChange** Nothing changes
- **Delete** Unmanaged and deleted	v **Detach** Unmanaged, stays in Azure	! **Unsupported** Can't be evaluated

Stacks What-If

```
~ .../storageAccounts/stappb5plrxcdyi6my [2023-05-01]
= Management Status: "managed"
= Deny Status: "denyDelete"
~ sku.name: "Standard_LRS" => "Standard_GRS"
```

NIC26

Declarative deletion

Take a resource out of the template, and the stack deletes it.



The template is the source of truth

It helps you manage drift.

Detach instead

DetachAll leaves it in Azure, unmanaged.

Demo

Take the network out of the template

Stacks What-If in a pipeline



Regular What-If vs Stacks What-If

	Regular What-If	Stacks What-If
Works with	Regular deployments	Deployment stacks
The result	Output	Resource
Noise	Included	Filtered
Removed resources	Ignores	Shows Detach or Delete
Deny settings	Not shown	Shown on every resource

Our recommendation: use Stacks What-If, and move to stacks to get it.

NIC26

PART 2

Debugging

NIC26

Catching problems before you deploy

New

1 Lint	2 Snapshot	3 Stacks What-If	4 Deploy
In VS Code, and bicep lint in CI Unused params, a missing @secure(), and rules in bicepconfig.json	**bicep snapshot in the PR** Exactly what changed in the template. Runs locally and doesn't deploy anything	**Right before you deploy** Exactly what Azure will do, including Stacks changes such as detach and delete	**Via the stack** Deny settings block changes that aren't done via the stack

On your machine

Checks what's live in Azure

NIC26

Snapshot vs What-If

`bicep snapshot`

What changed in my template?

Runs locally and gives the same answer every time, so it fits a PR check.

`Stacks What-If`

What will Azure actually do?

Runs in Azure against the real resources, and knows about detach and delete.

bicep snapshot --mode validate

```
- [resourceId(..., 'vnet-app')]
    type: "Microsoft.Network/virtualNetworks"
    name: "vnet-app"
```

Stacks What-If

```
- .../virtualNetworks/vnet-app
~ Management Status: "managed" => "notManaged"
~ Deny Status: "denyDelete" => "none"
```

NIC26

Demo

Snapshot and lint on a pull request

```
PS> bicep snapshot main.bicepparam --mode overwrite
PS> Copy-Item main.refactor.bicep main.bicep   # the refactor
PS> bicep snapshot main.bicepparam --mode validate   # passes
PS> Copy-Item main.v2.bicep main.bicep   # the pull request
PS> bicep snapshot main.bicepparam --mode validate
Snapshot validation failed. Expected no changes, but found ...
```

NIC26

PART 3

AI-assisted Bicep

Moving what you already have to stacks

NIC26

Prompting

Prompt from the demo

The resources in **rg-demo-nic2026-brownfield were built by hand in the portal** 1. Bring them under a new deployment stack called **stack-brownfield** 2 **without changing anything** 3. Put the template in a **brownfield folder** 4 and **show me the Stacks What-If before anything gets deployed** 5.

1 What's there now	2 The exact names	3 What can't change	4 Where the files go	5 Where to stop

In the skill, use it and iterate it every time

Run bicep lint. Run Stacks What-If.

Sanity checking New

bicep snapshot before, validate after, to ensure no resource changes.

NIC26

Demo

AI-assisted Azure Resources to a Stack



NIC26

Try it out

1

New

Add a snapshot check to your PRs

Commit the snapshot file, and run validate mode in CI.

`aka.ms/bicep/snapshot`

2

Run Stacks What-If on one of your stacks

If it hasn't been deployed since August, deploy it once to get a baseline.

`aka.ms/stackswhatif`

3

Write a skill for a job you do frequently

Check AI-authored Bicep with Stacks What-If.

VS Code docs: Agent Skills

NIC26

Q&A

Book time with me (or grab a coffee ☺)

If you want to talk through moving your own setup to stacks, pick a time here at NIC or after.

aka.ms/bicep/NIC26/meet

Slides and demo files	aka.ms/bicep/NIC26/demos
Stacks What-If docs	aka.ms/stackswhatif
GA announcement	aka.ms/stackswhatifGA
bicep snapshot	aka.ms/bicep/snapshot
Bicep MCP server	aka.ms/bicep/mcp
Feedback and issues	aka.ms/stacks-feedback
Monthly community call	aka.ms/armnews

Thanks for coming, and please rate this session.

NIC26